Formula sheet — Assignment 1

1 Geometry and bounding box

$$\text{Sphere } \mathcal{S}: \quad x^2 + y^2 + z^2 \le k^2 \tag{1.1}$$

$$\text{Torus } \mathcal{T}: \quad \left(\sqrt{(x-x_c)^2 + (y-y_c)^2} - R\right)^2 + (z-z_c)^2 \le r^2 \tag{1.2}$$

$$A = \mathcal{S} \cap \mathcal{T}, \qquad I(\mathbf{x}) = \begin{cases} 1, & \mathbf{x} \in A \\ 0, & \mathbf{x} \notin A \end{cases}, \qquad \mathbf{x} = (x, y, z) \tag{1.3}$$

$$\mathbf{x} \in A \;\Rightarrow\; |x| \le k, \quad |y| \le k, \quad z_c - r \le z \le z_c + r \tag{1.4}$$

$$\text{Bounding box:} \quad A \subset B = [x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [z_{\min}, z_{\max}] \tag{1.5}$$

$$\text{Volume of } B: \quad V_B = (x_{\max} - x_{\min})(y_{\max} - y_{\min})(z_{\max} - z_{\min}) \tag{1.6}$$

2 From integral to expectation

$$\mathbb{E}\big[g(\mathbf{X})\big] = \int g(\mathbf{x})\, f_{\mathbf{X}}(\mathbf{x})\, d\mathbf{x} \tag{2.1}$$

$$\mathbf{X} \sim U(B): \qquad f_{\mathbf{X}}(\mathbf{x}) = \begin{cases} \dfrac{1}{V_B}, & \mathbf{x} \in B \\ 0, & \text{otherwise} \end{cases} \tag{2.2}$$

$$V = \iiint_B I(x, y, z)\, dx\, dy\, dz = V_B\, \mathbb{E}\big[I(\mathbf{X})\big] \tag{2.3}$$

3 Monte Carlo estimator

$$x = x_{\min} + (x_{\max} - x_{\min})\, U, \qquad U \sim U(0,1) \quad \text{(same for } y, z) \tag{3.1}$$

$$\hat{V}_n = V_B\, \frac{H}{n}, \qquad H = \sum_{i=1}^{n} I(\mathbf{X}_i) \tag{3.2}$$

$$\text{Bias}(\hat{V}_n) = \mathbb{E}\big[\hat{V}_n\big] - V = 0 \tag{3.3}$$

4 Law of large numbers

$$\text{WLLN:} \quad \mathbb{P}\Big(|\bar{X}_n - \mu| > \varepsilon\Big) \xrightarrow{n\to\infty} 0 \quad \forall\, \varepsilon > 0 \tag{4.1}$$

$$\text{SLLN:} \quad \mathbb{P}\Big(\lim_{n\to\infty} \bar{X}_n = \mu\Big) = 1 \tag{4.2}$$

$$\hat{V}_n = V_B\, \frac{H}{n} \;\xrightarrow{n\to\infty}\; V_B\, \mathbb{E}\big[I(\mathbf{X})\big] = V \tag{4.3}$$

5 Assumptions

$$\mathbf{X}_i \sim U(B) \tag{A1}$$

$$\mathbf{X}_i \perp\!\!\!\perp \mathbf{X}_j, \quad i \ne j \tag{A2}$$

6 Central limit theorem and error

$$\mathrm{Var}(X_1 + \cdots + X_n) = \mathrm{Var}(X_1) + \cdots + \mathrm{Var}(X_n) = n\sigma^2, \qquad \sqrt{n\sigma^2} = \sigma\sqrt{n} \tag{6.1}$$

$$\lim_{n\to\infty} \mathbb{P}\left(\frac{(X_1 + \cdots + X_n) - n\mu}{\sigma\sqrt{n}} < z \right) = \Phi(z), \qquad z = \frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \tag{6.2}$$

$$f = \frac{V}{V_B}, \qquad \sigma^2 = V_B^2\, f(1-f), \qquad \mathrm{SE} = V_B \sqrt{\frac{\hat{f}(1-\hat{f})}{n}} \;\propto\; \frac{1}{\sqrt{n}} \tag{6.3}$$

$$\text{CI:} \quad \hat{V}_n \pm z\,\mathrm{SE} \tag{6.4}$$

7 Reference volume

$$V = \int A_{\mathrm{ring}}(z)\, dz, \qquad A_{\mathrm{ring}}(z) = \pi\Big(\rho_{\mathrm{out}}(z)^2 - \rho_{\mathrm{in}}(z)^2\Big) \tag{7.1}$$

$$\text{Trapezoid: error } O\Big(n^{-2}\Big), \;\; O\Big(n^d\Big) \text{ evaluations} \qquad \text{MC: error } O\Big(n^{-1/2}\Big) \;\; \forall\, d \tag{7.2}$$

$$R + r \le k \;\Rightarrow\; \mathcal{T} \subset \mathcal{S} \;\Rightarrow\; V = (2\pi R)(\pi r^2) = 2\pi^2 R r^2 \tag{7.3}$$

8 Logistic map

$$X_{k+1} = m\, X_k (1 - X_k) \tag{8.1}$$

$$X_k \in \left[\frac{m^2(4-m)}{16},\; \frac{m}{4} \right] \tag{8.2}$$

$$X_k \nsim U(0,1) \;\; \text{(A1 fails)}, \qquad X_{k+1} \text{ fixed by } X_k \;\; \text{(A2 fails)} \quad \Rightarrow \quad \mathrm{Bias}(\hat{V}_n) \neq 0 \tag{8.3}$$

9 Mixture proposal

$$q(\mathbf{x}) = \begin{cases} \dfrac{p}{V_B} + \dfrac{1-p}{V_S}, & \mathbf{x} \in S \\ \dfrac{p}{V_B}, & \mathbf{x} \in B,\; \mathbf{x} \notin S \end{cases} \qquad A \subset S \subset B \tag{9.1}$$

$$\mathbb{E}\Big[\hat{V}_{\mathrm{naive}}\Big] = \mathbb{E}\Big[V_B\, \frac{H}{n}\Big] = V\Big(p + (1-p)\,\frac{V_B}{V_S}\Big) \neq V \tag{9.2}$$

$$\hat{V}_w = \frac{1}{n}\sum_{i=1}^{n} \frac{I(\mathbf{X}_i)}{q(\mathbf{X}_i)}, \qquad \mathbb{E}\Big[\hat{V}_w\Big] = V \tag{9.3}$$

$$\mathrm{Var}\left(\hat{V}_w\right) = \frac{1}{n}\Big(\frac{V}{q_S} - V^2\Big), \qquad q_S = \frac{p}{V_B} + \frac{1-p}{V_S} \tag{9.4}$$